

Training and Evaluation Results Discussion

Jorge Peguero – W214777728

ITAI 1378

This document analyzes the training and evaluation results of the YOLOv8n-cls model trained to classify video frames into two categories: fire and nofire. The model was trained for 20 epochs on the Wildfire Dataset.

Training Results:

(training setup discussed in Yolo_TrainingAndEvaluation_Jorge_Peguero_ITAI1378.pdf)

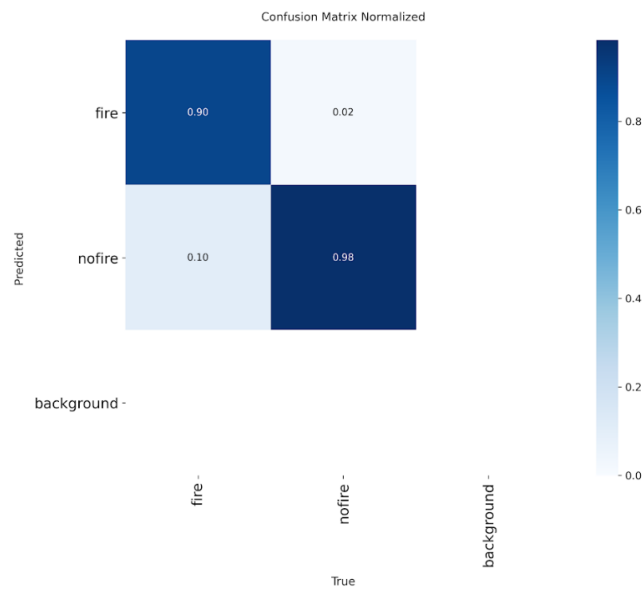
```
EarlyStopping: Training stopped early as no improvement observed in last 5 epochs. Best results observed at epoch 13, best model saved as best.pt.
To update EarlyStopping(patience=5) pass a new patience value, i.e. 'patience=300' or use 'patience=0' to disable EarlyStopping.

18 epochs completed in 5.127 hours.
```

The training stopped at epoch 18 due to the patience parameter set to 5. The model took 5.127 hours to converge, and the best epoch was number 13.

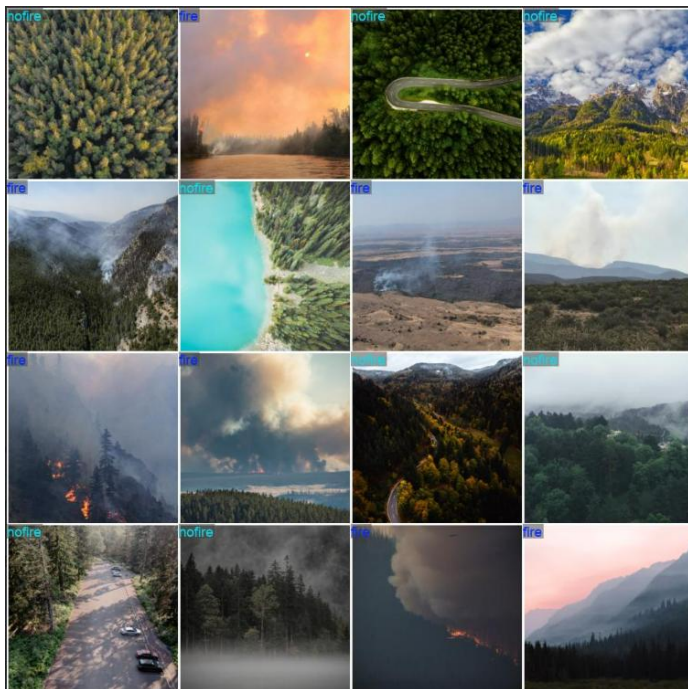
A	B	C	D	E	F	G	H	I
epoch	time	train/loss	metrics/accuracy_to	metrics/accuracy_to	val/loss	lr/pg0	lr/pg1	lr/pg2
1	1016.76	0.49793	0.87811	1	0.29849	0.000550958	0.000550958	0.000550958
2	2022.89	0.29264	0.88557	1	0.31278	0.00105185	0.00105185	0.00105185
3	3040.3	0.24728	0.91542	1	0.27647	0.00149772	0.00149772	0.00149772
4	4053.18	0.20539	0.90547	1	0.25518	0.00141945	0.00141945	0.00141945
5	5062.7	0.20515	0.91791	1	0.24645	0.00133693	0.00133693	0.00133693
6	6071.65	0.18399	0.9204	1	0.25542	0.00125442	0.00125442	0.00125442
7	7084.8	0.15288	0.9204	1	0.27812	0.0011719	0.0011719	0.0011719
8	8100.55	0.14729	0.9204	1	0.26265	0.00108938	0.00108938	0.00108938
9	9155.09	0.13951	0.9403	1	0.28778	0.00100687	0.00100687	0.00100687
10	10185.3	0.12997	0.92786	1	0.27587	0.000924352	0.000924352	0.000924352
11	11203.9	0.1101	0.93532	1	0.25286	0.000841835	0.000841835	0.000841835
12	12241.2	0.10804	0.92786	1	0.28904	0.000759318	0.000759318	0.000759318
13	13280.5	0.09303	0.95025	1	0.23674	0.000676802	0.000676802	0.000676802
14	14314.8	0.06683	0.94279	1	0.2331	0.000594286	0.000594286	0.000594286
15	15357	0.07488	0.9403	1	0.25437	0.000511769	0.000511769	0.000511769
16	16395.5	0.07689	0.93781	1	0.22543	0.000429253	0.000429253	0.000429253
17	17430.6	0.08324	0.93781	1	0.25676	0.000346736	0.000346736	0.000346736
18	18455.4	0.06428	0.92786	1	0.25355	0.00026422	0.00026422	0.00026422

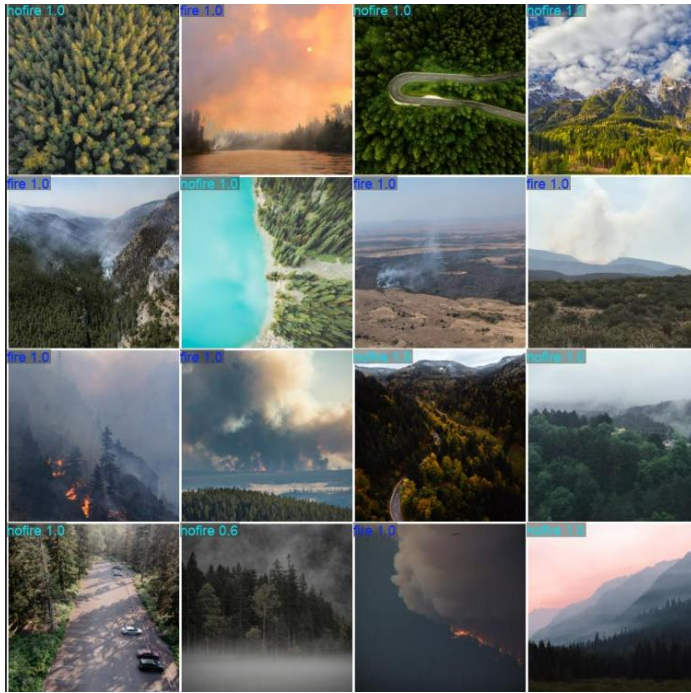
This are the results from every epoch. Epoch 13 had the best accuracy, as well as one of the lowest train and validation loss.



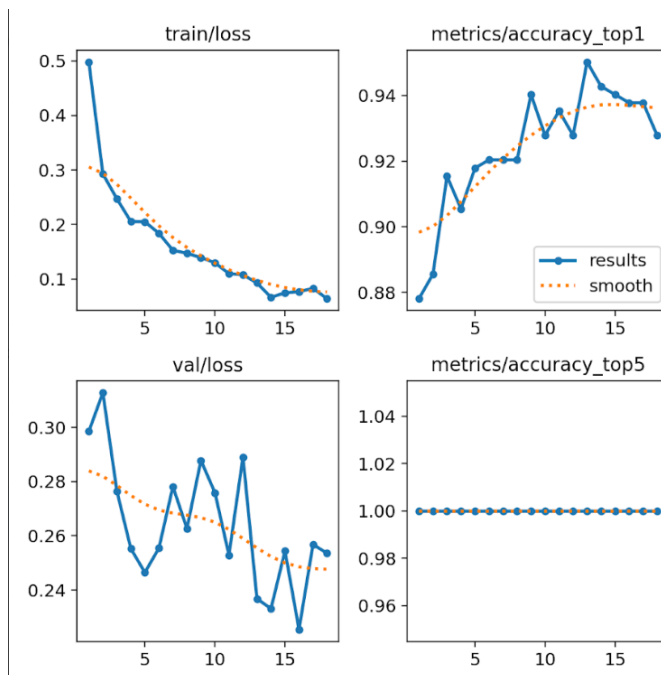
The model reached a 0.9 recall.

Some examples of predictions in training are:





The model is always very confident about its predictions, and in this example only shows an error on the bottom right corner. Probably the model confused the smoke with fog.

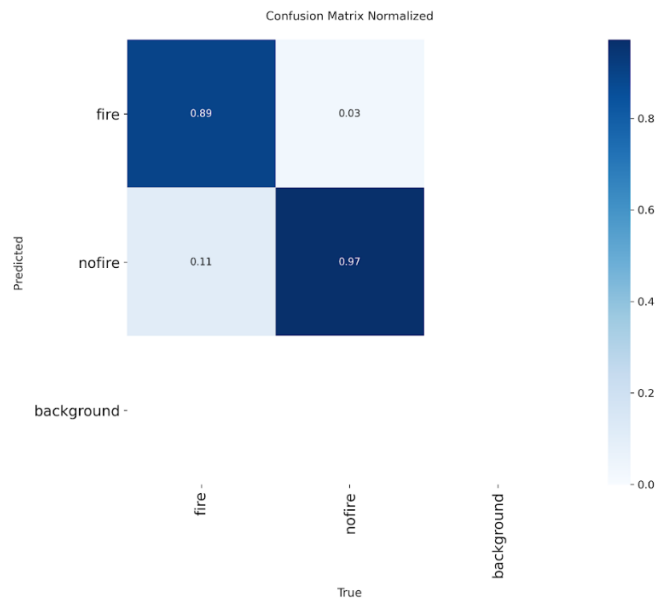


Top5 accuracy is not applicable in this project, since we only have 2 classes. But the tendency of train loss and validation loss was to go down. On the other hand, the tendency of accuracy was to go up.

Testing Results:

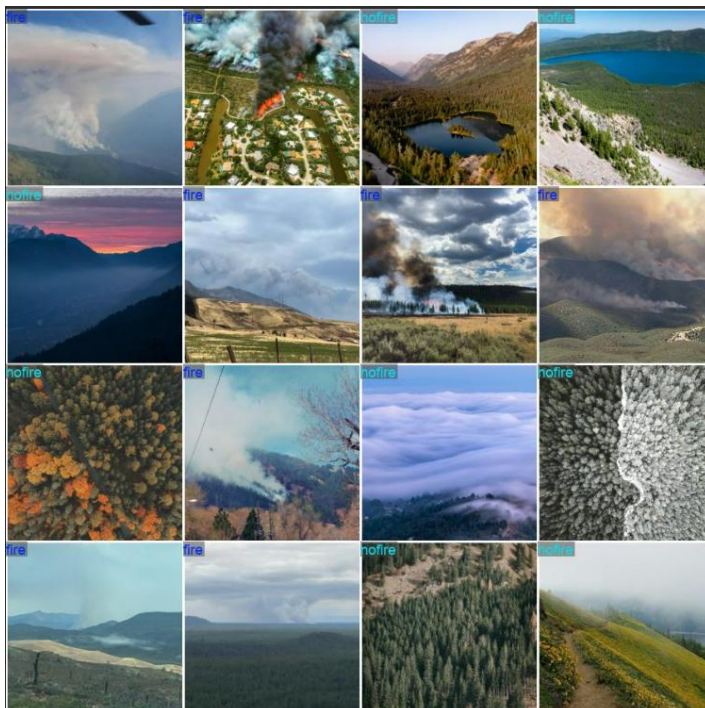
(test setup discussed in Yolo_TrainingAndEvaluation_Jorge_Peguero_ITAI1378.pdf)

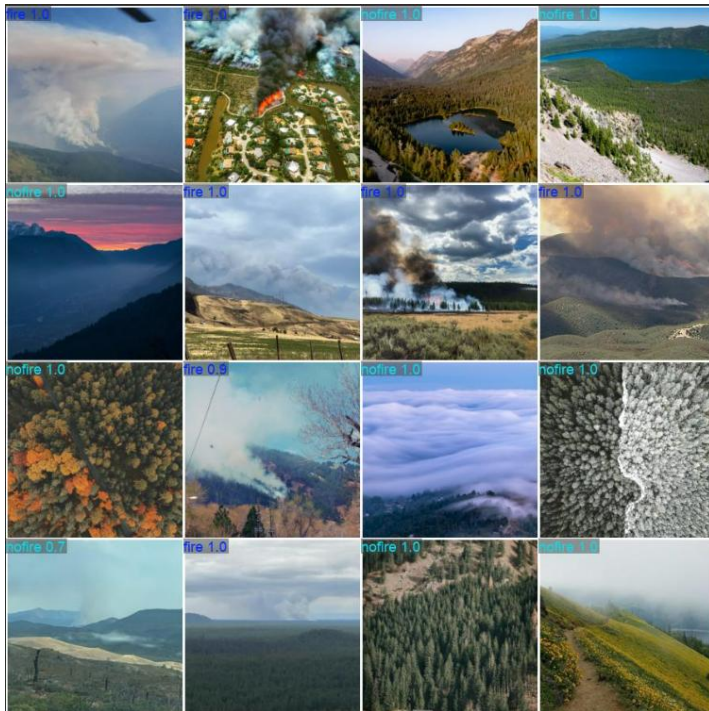
The model was evaluated on the test split, never used in training.



The model reached a 0.89 recall, consistent with the results on training. It is normal that in new data the metrics go down a little bit.

Some examples of predictions in final evaluation are:





Again high confidence in its predictions and one error at the bottom left. Probably the model confused the smoke with the sky because of the similar color they have.

The model is performing well, and the results were consistent between train and test.